

Basic CLI Practice Assignments

Windows Command Prompt (CMD) • Beginner Level • 5 Practical Problems

These assignments are based on the commands practiced in class: **dir**, **cd**, **cd ..**, **cd .**, **mkdir**, **echo**, **> (redirection)**, **del**, **rmdir**, **rmdir /s**, and **rmdir /s /q**. The class practice demonstrated navigating directories, creating folders, creating a text file with echo, deleting files, and removing non-empty directories.

Source basis: the uploaded class practice shows commands such as **cd**, **dir**, **mkdir**, **echo ... > file.txt**, **del**, and **rmdir /s /q**.
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Assignment 1 — Create and Explore a Student Folder

Difficulty	Easy
Commands/Concepts	mkdir, cd, cd .., dir

Problem Description

Create a folder named **student_cli** on the Desktop. Enter the folder and verify that it is empty. Then move back to the Desktop and verify that the folder exists.

Input / Commands to Perform

```
1. mkdir student_cli
2. cd student_cli
3. dir
4. cd ..
5. dir
```

Expected Output / Result

After entering **student_cli**, the directory listing should show **0 File(s)** and the folder should contain no student-created files. After returning to Desktop, **student_cli** should appear in the directory listing.

Additional Requirement

Do not use File Explorer. Perform the complete task using CMD only.

Assignment 2 — Create a Notes File Using CLI

Difficulty	Easy
Commands/Concepts	mkdir, cd, echo, >, dir, del

Problem Description

Inside a new folder named **cli_notes**, create a text file named **notes.txt**. Store the sentence **CLI makes computer operation easy** inside the file. Verify the file using **dir** and then delete the file.

Input / Commands to Perform

```
1. mkdir cli_notes
2. cd cli_notes
3. echo CLI makes computer operation easy > notes.txt
4. dir
5. del notes.txt
6. dir
```

Expected Output / Result

The first **dir** after creating the file should show **notes.txt**. The final **dir** should show **0 File(s)** because the file has been deleted.

Additional Requirement

The file must be created through command-line redirection, not a text editor.

Assignment 3 — Build a Small Project Structure

Difficulty	Medium
Commands/Concepts	mkdir, cd, cd .., dir

Problem Description

Create a folder named **my_project**. Inside it create three folders: **src**, **docs**, and **test**. Enter each folder at least once using **cd** and return to the project root using **cd ..**. Finally use **dir** to verify the three folders.

Input / Commands to Perform

```
1. mkdir my_project
2. cd my_project
3. mkdir src
4. mkdir docs
5. mkdir test
6. dir
```

Expected Output / Result

The final directory listing inside **my_project** should contain the three directories: **src**, **docs**, and **test**, with no student-created files required.

Additional Requirement

After creating the structure, demonstrate navigation into **src**, **docs**, and **test** and back to **my_project**.

Assignment 4 — Project Files and Cleanup

Difficulty	Medium
Commands/Concepts	mkdir, cd, cd .., echo, >, dir, del

Problem Description

Create a folder named **practice_project** with two subfolders: **src** and **docs**. Inside **src**, create **app.txt** containing **Application Code**. Inside **docs**, create **readme.txt** containing **Project Documentation**. Verify both files using **dir**. Then delete both files and verify that both folders are empty.

Input / Commands to Perform

```
1. mkdir practice_project
2. cd practice_project
3. mkdir src
4. mkdir docs
5. cd src
6. echo Application Code > app.txt
7. dir
8. cd ..
9. cd docs
10. echo Project Documentation > readme.txt
11. dir
```

Expected Output / Result

src should contain **app.txt** and **docs** should contain **readme.txt**. After using **del app.txt** and **del readme.txt**, a final **dir** in each folder should show **0 File(s)**.

Additional Requirement

Keep the folder structure intact; only the two files should be deleted.

Assignment 5 — Understand and Remove a Non-Empty Directory

Difficulty	Hard
Commands/Concepts	mkdir, cd, echo, >, dir, rmdir, rmdir /s /q

Problem Description

Create a folder named **cli_cleanup** and inside it create a file named **data.txt** with the text **Temporary Data**. Move back to Desktop and try to remove **cli_cleanup** using **rmdir cli_cleanup**. Observe the result because the directory is not empty. Then remove the complete directory and its contents using **rmdir /s /q cli_cleanup**. Finally use **dir** to confirm that the folder no longer exists.

Input / Commands to Perform

```
1. mkdir cli_cleanup
2. cd cli_cleanup
3. echo Temporary Data > data.txt
4. cd ..
5. rmdir cli_cleanup
6. rmdir /s /q cli_cleanup
7. dir
```

Expected Output / Result

The first **rmdir cli_cleanup** should fail because the directory contains **data.txt**, producing a message that the directory is not empty. The **rmdir /s /q cli_cleanup** command should then remove the directory and its contents without asking for confirmation. The final **dir** should not list **cli_cleanup**.

Additional Requirement

Run the commands in the given order and explain in one or two sentences why the first **rmdir** fails.

Student Submission Checklist

- Complete all 5 assignments using Windows Command Prompt.
- Do not use File Explorer for creating, deleting, or navigating folders/files.
- Take a screenshot of the important command/result for each assignment.
- Write the commands you used below each assignment in your submission.
- For Assignment 5, explain why rmdir fails when the directory is not empty.

Trainer Note: The assignments intentionally stay within the commands demonstrated in the provided class practice. The uploaded practice includes examples of directory navigation, mkdir, echo with file redirection, del, and the difference between rmdir and rmdir /s /q. ■filecite■turn1file1■L69-L106■